

ANKIT MATHANKER

ankitma@umich.edu — University of Michigan, Ann Arbor, USA

Website: <https://ankitmathanker.github.io/>

Summary

Ph.D. Candidate in Chemical Engineering with research experience in atomistic modeling, reaction kinetics, machine learning, interatomic potentials, generative modeling, and agentic AI for applications in material science for energy and sustainability.

Education

- **Ph.D. in Chemical Engineering and Scientific Computing** Aug 2021–Present
University of Michigan, USA GPA: 4.0/4.0
 - **Master of Science in Chemical Engineering** Jan 2018–Jan 2020
University of Alberta, Canada GPA: 3.8/4.0
 - **Bachelor of Technology in Chemical Engineering** Jul 2013–May 2017
IIT (ISM) Dhanbad, India GPA: 8.9/10.0
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Skills

- **Atomistic Modeling & Computational Chemistry:** Density Functional Theory, Atomic Simulation Environment, Pymatgen, Vienna Ab initio Simulation Package, JDFTx, GROMACS, LAMMPS (MD/MC), transition-state search (NEB, vibrational analysis), MLIP-based molecular dynamics workflows, microkinetic modeling
 - **Scientific Machine Learning & AI:** Python (NumPy, Pandas, SciPy, scikit-learn), PyTorch, MACE, DeepMD, uncertainty quantification and calibration, scientific machine learning, Agentic workflows, OpenAI SDK, LLM-based multi-agent systems, workflow automation, active learning and Bayesian optimization.
 - **High-Throughput Computing & Infrastructure:** HPC environments (Slurm, PBS), cluster job submission and orchestration, high-throughput simulation pipelines, GPU computing (CUDA)
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Research Experiences

- **Graduate Research Assistant**, University of Michigan, USA Aug 2021–Present
 - Understanding the effect of electrode potential, electrolyte species, and co-reactants on the electrocatalytic hydrogenation of organics using density functional theory, molecular dynamics.
 - Developing an agentic AI workflow for autonomous transition-state discovery in quantum chemistry simulations, combining multi-agent planning, workflow automation, DFT-based mechanistic modeling. (Ongoing)
- **Graduate Intern**, Lawrence Livermore National Laboratory, USA Jun–Aug 2024
 - Estimating potential-dependent physicochemical properties at metal–electrolyte interfaces using machine learning interatomic potentials developed using MACE architecture.
- **Graduate Research Assistant/Research Assistant**, University of Alberta, Canada Jan 2018–Jan 2021
 - Investigate the synergistic impact of agricultural and forest residue on hydrothermal liquefaction using supercritical water medium in an autoclave reactor.
- **Research Assistant (Indian Academy of Science Fellow)**, IIT Guwahati, India May–Jul 2016
 - Investigate the synthesis of low-cost tubular ceramic membrane (TCM) via the extrusion process. Experimentally and analytically measure water flux and porosity.
- **Undergraduate Dissertation**, IIT (ISM) Dhanbad, India May 2016–Apr 2017

- Synthesize PEI-impregnated adsorbents for CO₂ capture and simulate adsorbent packed bed for CO₂ adsorption on Aspen Adsorption.
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Publications

1. Madhavan, V.; Mathanker, A.; Sweeney, D.; Oohiro, T.; Goldsmith, B. R. TSAgent: An Agentic Workflow for Autonomous Transition State Search. <https://arxiv.org/abs/2605.14154>, (In review 2026).
 2. Mathanker, A.; Guo, J.; Goldsmith, B. R.; Varley, J.; Govindarajan, N. Estimating potential-dependent physicochemical properties at metal–electrolyte interfaces using machine learning interatomic potentials. *ACS Electrochemistry*, in press 2025-2026, doi.org/10.1021/acselectrochem.5c00540.
 3. de Kam, L. B. T.; Zhu, J. X.; Mathanker, A.; Doblhoff-Dier, K.; Govindarajan, N. Benchmarking short-range machine learning potentials for atomistic simulations of metal/electrolyte interfaces. <https://arxiv.org/abs/2602.22931>, (In Review *J. Chem. Phys.*).
 4. Mathanker, A.; Sharma, G.; Tran, B.; Singh, N.; Goldsmith, B. R. Effect of ions on the aqueous-phase adsorption of small aromatic organics on silver. *J. Phys. Chem. C* 2025, 129, 29, 13433-13444.
 5. Mathanker, A.; Halarnkar, S.; Tran, B.; Singh, N.; Goldsmith, B. R. Synergistic effects in organic mixtures for enhanced catalytic hydrogenation and hydrodeoxygenation. *Chem Catalysis* 2024, 4, 101135.
 6. Mathanker, A.; Yu, W.; Singh, N.; Goldsmith, B.R. Effects of ions on electrocatalytic hydrogenation and oxidation of organics in aqueous phase. *Curr. Opin. Electrochem.* 40, 101347 (2023).
 7. Das, S.; Mathanker, A.; Pudasainee, D.; Khan M.; Kumar, A.; Gupta, R. Synergistic effect of water and co-solvents on the hydrothermal liquefaction of agricultural biomass to produce heavy oil. *International Journal of Energy for a Clean Environment* 2022, 23(4):31-45.
 8. Mathanker, A.; Das, S.; Pudasainee, D.; Khan, M.; Gupta, R. A review on hydrothermal liquefaction of biomass for biofuels production with special focus on the effect of process parameters, co-solvents and extraction solvents. *Energies* 2021, 14, 4916.
 9. Mathanker, A.; Pudasainee, D.; Kumar, A.; Gupta, R. Hydrothermal liquefaction of lignocellulosic biomass feedstock to produce biofuels: Parameter study and products characterization. *Fuel* 2020, 271, 117534.
 10. Mathanker, A. Hydrothermal liquefaction of lignocellulosic biomass to produce biofuels. *Thesis* 2020.
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Conference Talks and Posters (most recent)

- Mathanker, A. "Estimating potential-dependent physicochemical properties at metal–electrolyte interface using machine learning interatomic potentials" Gordon Research Conference - AI for Materials, Energy, and Chemical Science, 2026. Texas, USA.
 - Mathanker, A.; Guo, J.; Goldsmith, B. R.; Varley, J. B.; Govindarajan, N. "Estimating Water Coverage and Adsorption Isotherms at Electrolyte/Metal Interfaces Using Machine Learning Potentials". AIChE Annual Meeting: Catalysis and Reaction Engineering Division, 2025. Boston, Massachusetts, USA.
 - Mathanker, A.; Sharma, G.; Tran, B.; Singh, N.; Goldsmith, B. R. "Effect of ions on the aqueous-phase adsorption of small aromatic organics on silver". AIChE Annual Meeting: Catalysis and Reaction Engineering Division, 2025. Boston, Massachusetts, USA.
 - Mathanker, A. "Can aqueous ions modify the adsorption of organics on Ag?" The Student and Postdoc Summer Seminar Series, CSiDIR, 2025, University of Michigan, Ann Arbor, MI, USA.
 - Mathanker, A.; Sharma, G.; Tran, B.; Singh, N.; Goldsmith, B. R. "Effect of ions on the aqueous-phase adsorption of small aromatic organics on Ag". The 46th Michigan Catalysis Society Symposium, 2025. Warren, Michigan, USA.
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Teaching Experiences

- **Graduate Teaching Assistant**, University of Michigan, USA 2023
 - CHE 538: Statistical thermodynamics
 - Managed a class of 57 students, facilitated collaborative learning via Piazza discussions and office hours, and developed additional resources to address common challenges faced by students from diverse departments.
 - Applied pedagogical training in course material development, student engagement, and feedback integration, leading to increased student collaboration and engagement throughout the semester.
 - **Graduate Teaching Assistant**, University of Alberta, Canada 2019
 - CHE 316: Equilibrium stage processes, Spring and Fall terms.
 - Developed course tutorials and delivered over 40 hours of seminars to a class of 42 students and incorporated classroom engagement techniques such as role reversal for active and collaborative learning.
 - **Graduate Teaching and Learning (Foundation and Practicum)**, University of Alberta, Canada 2018–2019
 - Advanced foundational teaching skills and strategies for effective classroom engagement through workshops on course design and pedagogy.
 - Gained hands-on experience in the advanced teaching practices, learning objectives, micro teaching skills and lesson planning strategies.
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Awards

- LLM Hackathon for Applications in Materials and Chemistry–Visionary Awards for **RedoxFlow**
 - Awarded to top 25 teams for outstanding vision for the use of Agentic-AI in field of material science.
 - CRE Division Travel Award-AIChE AICHE Annual Meeting, USA
 - CRE Travel Award is presented by the AIChE's Catalysis and Reaction Engineering Division in recognition of exceptional individuals who have made significant advancements in the field.
 - Rackham Travel Award (2022,2023,2024) University of Michigan, USA
 - Awarded to present research at national and international conferences.
 - Captain Thomas Farrell Greenhalgh Memorial Graduate Scholarship University of Alberta, Canada
 - Awarded for the excellent academic achievements to top 5 percent of incoming cohort.
 - Mary Louise Imrie Graduate Student Award University of Alberta, Canada
 - Awarded to present extraordinary research work in prestigious conferences across the globe.
 - Graduate Student Association Travel Award University of Alberta, Canada
 - Awarded to present graduate research in international conferences.
 - Summer Research Fellowship Programme Scholarship Indian Academy of Science, Bengaluru, India
 - Awarded to top 10 percent applicants through Indian Academy of Sciences to conduct academic research internship in top programs over summer.
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Industrial Work Experience

- **Management trainee**, Polyplex Corporation Ltd., India Jul–Sept 2017
 - Trained in end-to-end production process handling, quality assurance for PET and PE polymer chips and packaging film manufacturing.
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Academic/ Non-academic Community services

- Graduate Student Director, AIChE Catalysis and Reaction Engineering Division, 2025-2026
 - Lead the management of the social media team with Prof. Omar Abdelrahman and spearheaded initiatives to enhance graduate student engagement and participation in division events.

- Graduate Student Representative, Michigan Chapter of North American Catalysis Society, 2022-2024
 - Promoted graduate student perspectives, facilitating greater involvement and collaboration within the society.
- Graduate Recruitment Planning Member, Department of Chemical Engineering, University of Michigan, 2022-2024
 - Organized and coordinated recruitment activities, including technical sessions and social events, for incoming graduate students.
- Core Team Member, Chemical Engineering Society (ChEGs), University of Michigan, 2022-2023
 - Helped in organization of various academic and social events for graduate students, fostering a strong community within the department.
- Vice President, Toastmasters International, University of Alberta, Canada, 2018-2019
 - Directed membership management, public relations efforts, and organized public speaking competitions to enhance communication skills within the university community.
- Volunteer, UNITEA – Community Social Work Team, University of Alberta, Canada, 2018-2019
 - Collaborated with the community welfare team to promote mental health and well-being initiatives among graduate students.
- Volunteer, Kartavya – Non-Governmental Organization, IIT (ISM) Dhanbad, India, 2015-2017
 - Tutored approximately 20 underprivileged students weekly, providing over 200 hours of academic support while leading a team of 50 undergraduate students to manage their school responsibilities.
- President, Chemical Engineering Society, IIT (ISM) Dhanbad, India, 2015-2016
 - Organized various academic and social events, including leading a team for the department's annual symposium, which brought together undergraduate and graduate students.